



PRODUCT NAME

Industrial-grade Dishwashing Liquid Formula

Description

Manual dishwashing detergent for kitchen use

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Sodium Lauryl Ether Sulfate (SLES) 70%	Sodium Laureth Sulfate	25%	Primary surfactant for cleaning and foaming
2	Cocoamidopropyl Betaine	Cocamidopropyl Betaine	10%	Foam booster and secondary surfactant
3	Sodium Chloride	Sodium Chloride	1.5%	Viscosity modifier
4	Sodium Citrate	Sodium Citrate	2%	Chelating agent
5	Ethanol	Ethanol	2%	Solvent and preservative support
6	Fragrance	Parfum	0.5%	Provides product scent
7	Preservative	Benzisothiazolinone	0.1%	Prevents microbial growth
8	Colorant	CI 42090	0.05%	Provides product color
9	Water	Aqua	58.85%	Solvent and formulation bulk
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Phase 7: Manufacturing Phase

Phase 8: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions